

CRISPR Multiplexing Optimizer

Project Definition Document

Version: 1.0 | **Date:** May 2026 | **Status:** Active Development

WHAT — Project Overview

Purpose

The CRISPR Multiplexing Optimizer is a computational tool that optimizes simultaneous CRISPR gene edits (10-50 sites) using machine learning predictions and delivery constraints. It addresses a critical gap in genome editing: no existing tool handles large-scale multiplexed CRISPR designs with ML-powered efficiency predictions.

Problem Statement

- Handle only 1-4 guide RNAs per design
- Lack ML-powered efficiency predictions
- Don't integrate delivery constraints (LNP, AAV, etc.)
- Missing off-target risk estimation

Solution

- ✓ ML-powered on-target efficiency scoring (Doench algorithm)
- ✓ Off-target risk estimation
- ✓ Multiplex delivery capacity constraints
- ✓ Composite ranking (efficiency × safety)
- ✓ Web interface + CLI

Target Users

- Academic researchers — Gene editing scientists
- Biotech companies — Therapeutic development teams
- iGEM teams — Synthetic biology competitions
- Contract research organizations — Drug discovery

HOW — Technical Implementation

Architecture

Three-layer architecture: User Interface → Core Optimizer → ML Models

Key Components

1. GuideRNA Class

```
@dataclass class GuideRNA: sequence: str # 20nt gRNA sequence target_gene: str
# Target gene name genomic_position: str # Chromosome:position # Properties
gc_content: float # GC percentage doench_score: float # On-target efficiency
(0-1) off_target_risk: float # Off-target risk (0-1)
```

Delivery Capacities

Method	Max Guides
LNP	50
Lentivirus	10
AAV	4
Lipofection	20
Electroporation	30

WHERE — Roadmap

Current State: v0.2.0 (Published)

✓ Doench scoring • ✓ Off-target risk • ✓ Validation data • ✓ Web UI • ✓ CLI

Phase 1: Core ML (Weeks 1-2)

- Training data pipeline (GUIDE-seq/CIRCLE-seq from GEO)
- Feature extraction (sequence + chromatin)
- ML model training (PyTorch DQN)

Phase 2: Validation (Weeks 3-4)

- Cross-validation (RMSE target: < 0.15)

- Benchmark publication (bioRxiv)
- Peer review

Phase 3: Production (Weeks 5-6)

- API deployment
- Benchling integration
- User authentication

Phase 4: Monetization (Weeks 7+)

- Academic (free) — Free tier
- Biotech (paid) — Fixed rate \$99/month
- Enterprise (custom) — Contact sales

SUCCESS METRICS

Metric	Target	Current
GitHub Stars	100+	New
Citations	10+	0
Accuracy (RMSE)	< 0.15	N/A
Active Users	100+	New
Revenue	\$10K ARR	\$0

RESOURCES

Repository: GitHub

Tech Stack: Python 3.10+, NumPy, Pandas, PyTorch, Streamlit, Click

Document generated by gstack workflow

Last updated: May 2026